

Tanya Saini

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Solution oriented Software Engineer with 5 years of full-stack experience, leading development efforts and building large-scale, secure, and scalable applications. Outcome driven Project Manager with 3 years of experience in problem solving, negotiation, resource management, leadership and team management that drove user engagement and business value for SaaS products. Strong background in conducting code and design reviews.

PROFESSIONAL EXPERIENCE

Software Project Manager | Software Engineer | OpenWater Software LLC, New York City

August 2022 – September 2025

- Demonstrated leadership by heading Team and Steering meetings, ensuring stakeholders and team members adhered to the project plan, and facilitating discussions on project progress and constraints.
- Engineered and fine-tuned ML models and LLMs (TensorFlow, PyTorch) for sentiment analysis and user behavior classification, generating real-time insights and improving personalization accuracy by 18%.
- Designed and deployed full-stack applications with React (frontend) and Python/Java (backend), optimizing API performance and frontend rendering, reducing latency by 30% and increasing user engagement by 25%.
- Integrated AI-driven analytics into web platforms, leveraging predictive models, advanced behavior analysis, and interactive visualizations to enable data-driven decision-making and actionable insights.
- Developed Lookups that query from an outside source (Salesforce, a Google Sheet, a CSV, etc.) in the in-house Integrations Hub.
- Deployed and engineered a customizable user profile authentication system for web applications in more than 20 projects.
- Implemented a scalable theming system using HTML, CSS variables, and Tailwind CSS, enabling light/dark mode with JavaScript. Reduced CSS redundancy by 35% and improved UI consistency.
- Implemented agile methodology improving team productivity by 30% for project completion and resulting in a 25% increase in client satisfaction.
- Leveraged JSON Web Token to build multiple APIs for the integrated applicant and judge portal, providing personalized user data.
- Spearheaded the integration of a new analytics dashboard within the OpenWater platform, enabling clients to track application metrics in real-time.
- Managed the successful launch of new platform features, contributing to a 15% increase in user engagement.
- Assisted in the development of SaaS products, focusing on user onboarding experiences that improved retention rates by 20%.
- Developed pushback connectors for system applications using third party databases (e.g.: Okta, OAuth, iMIS) for periodic queries.
- Successfully planned, monitored, and controlled project timelines and schedules of inter-departmental projects that led to successful completion of project phases in partnership with global organizations for their web applications before deadlines.

Software Engineer | OpenWater Software LLC, New York City

July 2020 – July 2022

- Planned, developed and maintained 40+ full-stack web development projects for clients, leveraging front-end technologies (HTML, CSS, JavaScript), alongside back-end tools (Python, SQL), to deliver scalable and user-centric solutions.
- Designed and implemented integrations between OpenWater and external authentication systems (e.g. OAuth, SSO) across 15+ client projects to enhance security and streamline user access.
- Collaborated on the optimization of system reliability, performance and scalability across all software layers by working on system bugs.
- Enhanced the deployment pipeline, reducing the average release cycle by 30%, from 10 days to 7 days, accelerating time-to-market for new features.
- Actively contributed to software release cycles in an Agile SDLC, ensuring code quality and system readiness for production deployments.

SKILLS

- Programming Skills: Python, Java, JavaScript, C++, SQL
- Cloud Technologies: AWS, MS Azure
- Web Technologies: React, Node.js, Angular, RESTful APIs, HTML5, CSS3, Bootstrap
- Databases: MySQL, NoSQL, MongoDB, ArangoDB
- Tools & Technologies: Git, VS Code, Linux, Jira, Kafka, Docker, Tableau, Kubernetes, NumPy, Pandas, Matplotlib, Spring Boot
- PM: Agile, Waterfall, Kanban, Scrum, Excel, Time Management, Project cost estimation, Risk identification

EDUCATION

New York University (NYU) | New York City, NY

May 2020

Master of Science in Computer Science

Amity University | Noida, India

May 2018

Bachelor of Technology in Computer Science and Engineering

PROJECTS

Big Data and Sentiment Analysis on Yelp Dataset (Python, Pandas, NumPy)

- Analyzed dataset provided by 'Yelp' containing data on Yelps' businesses, reviews and user data for 'Yelp Dataset Challenge'.
- Studied the data from 'Yelp' using python libraries such as Pandas, Seaborn and visualized it using Matplotlib to gain useful insights.
- Resolved questions based on business/user trends and sentiment analysis which were pre-listed during the initial phase of the project.

Twitter Data Mining and Analysis (Python, PySpark, MS Azure, MongoDB)

- Retrieved Twitter user data using Twitter API and python script which was run locally. Pushed the data onto MongoDB hosted on Azure.
- Analyzed the data which was stored over the cloud on MongoDB using PySpark to identify patterns on public opinion and trends.
- Produced useful charts and insights using the twitter data collected using python libraries such as NumPy, Seaborn and Pandas.

PUBICATIONS

- Application of Swarm Intelligence in Autonomous Vehicles for Obstacle Avoidance published in Springer, Singapore